

The DeFi Walkaway Test
Can Users Actually Leave?

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The problem

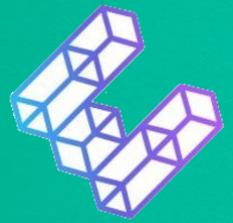
You deposit some money into a protocol

- Tomorrow
- The team disappears
- The website goes dark
- The API stops answering
- The indexer is gone

- The contracts are immutable and live forever

- Can you get your money out?



I tested it

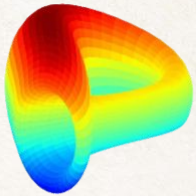















The Test

The walkaway test

- You have:
  - Your address
  - A node

- You don’t have:
  - Website
  - Any API
  - Indexer
  - Block Explorer
  - Docs
  - ABI

- Success is defined by getting your money back without anyone’s help/permission

How a run works

Who

- Top Ethereum protocols
- 100th largest holder of the main token
  - Per Ethplorer's top holders list
  - Nearest plain wallet if it's a contract
- A mainnet fork
- Same block for all

How

1. Find the position
2. Find the protocol's contracts
3. Learn the exit
4. Impersonate the wallet
5. Try it on the fork
6. Prove any blockers

DEV CON VIII INDIA



Meet the user

- 0xde804e46f67ebeecbe4c0a54efe3314 3cb7ab9e2
- 5084 stETH
- Plain wallet
- Deposited 5000 ETH in February 2026
- Block 26,248,199
- Developer who knows general Ethereum standards
- No protocol knowledge

Step 1
Gather information

- eth_getBalance
  - 406.1339 ETH
- eth_getTransactionCount
  - 36 nonce
- eth_getCode
  - None
- eth_getLogs
  - Transfer
  - TransferSingle
  - TransferBatch
  - From block 0 to fork
  - Transfers **to** and **from** the address
- name(), symbol(), decimals(), balanceOf(address)
  - All contracts

Contract	name()	symbol()
0xae7ab96520de3a18e5e111b5eaab095312d7fe84	Liquid staked Ether 2.0	stETH
0xeb9d5295194abb013b2d8b646f29dba833e726ad	coinsteth.net	claim rewards on coinsteth.net
0x93cba794cd4d48cfb62a5e435f8c843d12feee2a	stethaward.net	claim rewards on stethaward.net
0xd3cc310dd7b94d6ef06615f73a3029a67d935c5e	$ usd-coin.org	Visit https://usd-coin.org to claim rewa
0xc0076e208c5d9fb49634da0ad628674fb3c67587	# usd-coin.net	Visit https://usd-coin.net to claim rewa
0x5807dac5cc59f02dfa88154de7f9f2174c263814	$ wBTCLP.com	$ wBTCLP.com - Visit to claim bonus rewa
0x57b9d10157f66d8c00a815b5e289a152dedbe7ed	环球股	HQG
0x635701cc5fe41fe8abd02aa74beb02e3540e9bb2	# aBonusLP.com	Visit aBonusLP.com to claim $9283 in rew
0x2b4f9a8e992eb68e7e9ee4e49d961c2033e6b215	# apyeth.net	Visit https://apyeth.net to claim reward
0xa09a4e239fa3f604e23975622ba82798f30d393d	$ USDBonus.net	$ USDBonus.net - Visit to claim

```
❯ cast tx 0x5d78c88513b5f2d9f55c4fd9274ac003ad26117c45f9abcbbf5add36374f1383 --rpc-url https://l

blockHash            0x3ee12a8c9f9588157894e26fbca978c2dbe7fcb6f9ae75d40d76e3c6c0cd974e
blockNumber          24396338
from                 0xDE804e46f67ebeEcBe4c0a54Efe33143cb7ab9E2
transactionIndex     21
effectiveGasPrice    1657000000

accessList           []
chainId              1
gasLimit             182852
hash                 0x5d78c88513b5f2d9f55c4fd9274ac003ad26117c45f9abcbbf5add36374f1383
input                0xa1903eab000000000000000000000000011d0000000000000000000000000000000000011d0
maxFeePerGas         1657000000
maxPriorityFeePerGas 1657000000
nonce                35
r                    0xfd70cc052250256f333d28805491283ad1f0c9de8b86cd79ea1957b17971ec36
s                    0x57107823a2a84468989494f1f607cda0ba542593381519aa52599c05d47872d7
to                   0xae7ab96520DE3A18E5e111B5EaAb095312D7fE84
type                 2
value                500000000000000000000
yParity              0

blockTimestamp       1770363455
```

```
{
  "address": "0xae7ab96520de3a18e5e111b5eaab095312d7fe84",
  "blockHash": "0x3ee12a8c9f9588157894e26fbca978c2dbe7fcb6f9ae75d40d76e3c6c0cd974e",
  "blockNumber": "0x1744232",
  "blockTimestamp": "0x69859a3f",
  "data": "0x0000000000000000000000000000000000000000000010f0cf064dd591fffff",
  "logIndex": "0x1ce",
  "removed": false,
  "topics": [
    "0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef",
    "0x0000000000000000000000000000000000000000000000000000000000000000",
    "0x000000000000000000000000de804e46f67ebeecbe4c0a54efe33143cb7ab9e2"
  ],
  "transactionHash": "0x5d78c88513b5f2d9f55c4fd9274ac003ad26117c45f9abcbbf5add36374f1383",
  "transactionIndex": "0x15"
},
```

499999999999999999999 [4.999e21]

Sender

78 contracts found

- Exact symbol match "stETH"
- A contract can emit a Transfer event naming any address as the sender
  - Address poisoning

One incoming transfer

  - Transaction was sent by the user
  - Sent 5,000 ETH directly to the same stETH contract
  - Got 5,000 stETH back

```
CALL          user  -> 0xae7ab965…7fe84 (stETH)
  CALL          stETH -> 0xb8ffc3cd…88dc
    DELEGATECALL 0xb8ffc3cd… -> 0x2b33cf28…4425
  DELEGATECALL stETH -> 0x028271e3…cdb0
```

Step 1

```
00000041: PUSH4 0x48a0c8dd
00000046: EQ
00000047: PUSH2 0x0152
0000004a: JUMPI
0000004b: DUP1
```

- **debug_traceCall** with **callTracer**
- Real contract is on 0x028271e30a695c0527a0c50ca30603fed004cdb0
- **eth_getCode**
- **cast disassemble**
- Look for **PUSH4** that are followed by an **EQ**
- **86** candidates
- **9** matches **ERC-20**
- No matches on guess list

- 7 days window
- Get every **Transfer** emitted by the stETH contract in the window
- Check every address for code
- Group addresses by distinct senders
- Group addresses by amount sent

Step 2
Watch others

Top contract recipients by distinct senders (how many independent users send stETH there):

#	Contract	Code (B)	Senders	Transfers	stETH	%
1	`0x889edc2edab5f40e902b864ad4d7ade8e412f9b1`	2,497	**425**	616	63,592	15.09%
2	`0x7f39c581f595b53c5cb19bd0b3f8da6c935e2ca0`	6,492	195	746	56,624	13.44%
3	`0x9008d19f58aabd9ed0d60971565aa8510560ab41`	16,165	167	425	5,412	1.28%
4	`0x74de5d4fcbf63e00296fd95d33236b9794016631`	1,163	86	109	60	0.01%
5	`0x0cef320d0c5793ade37964e2661a033771e53cd5`	21,478	69	341	6,867	1.63%

Top contract recipients by amount:

#	Contract	Code (B)	Senders	Transfers	stETH	%
1	`0xe76c52750019b80b43e36df30bf4060eb73f573a`	2,497	2	16	76,174	**18.08%**
2	`0x889edc2edab5f40e902b864ad4d7ade8e412f9b1`	2,497	425	616	63,592	15.09%
3	`0x7f39c581f595b53c5cb19bd0b3f8da6c935e2ca0`	6,492	195	746	56,624	13.44%
4	`0x51c72848c68a965f66fa7a88855f9f7784502a7f`	20,805	11	171	19,571	4.65%
5	`0x53bcdcc18f3bada2cd749a72b28d55eb7abb3dd2`	9,343	2	39	15,290	3.63%

- 1,352 Mints
  - 54,088 stETH
  - 855 addresses
- 0 Burns

- 0x889edc2edab5f40e902b864ad4d7ade8e412f9b1
  - Has the most independent senders
  - 2nd most by amount

tx.to	Selector	Txs	Senders
0x889e…f9b1	0xacf41e4d	281	256
0x889e…f9b1	0xd6681042	117	115
0x889e…f9b1	0x7951b76f	32	29

Step 2
Watch others

- Looking at txs that use **0xacf41e4d**
- That looks like an NFT minting
  - Mint address
  - User address
  - ID
  - Transfer signature
- Send stETH -> get a numbered NFT

```
❯ cast keccak 'Transfer(address,address,uint256)'
0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef
❯ cast call 0x889edc2edab5f40e902b864ad4d7ade8e412f9b1 'ownerOf(uint256)(address)' 138796
0xb7Ab92766B16ba65C7a667AE021863fcC687c835
```


```
{
  "address": "0x889edc2edab5f40e902b864ad4d7ade8e412f9b1",
  "topics": [
    "0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef",
    "0x0000000000000000000000000000000000000000000000000000000000000000",
    "0x000000000000000000000000b7ab92766b16ba65c7a667ae021863fcc687c835",
    "0x000000000000000000000000000000000000000000000000000000000021e2c"
  ],
  "data": "0x",
  "blockHash": "0x2f03e331b21884f3c0f99a027a402d8d2d694a59261239261894b2823ac6d464",
  "blockNumber": "0x18fbf4b",
  "blockTimestamp": "0x6ad0c333",
  "transactionHash": "0xcfedd172e8e43de4dc82ddd2d6a194218936eb97fe402c7ff9dc9015afd5c87b",
  "transactionIndex": "0xfe",
  "logIndex": "0x459",
  "removed": false
}
```


Step 2
Watch others


```
{
  "address": "0x889edc2edab5f40e902b864ad4d7ade8e412f9b1",
  "topics": [
    "0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef",
    "0x0000000000000000000000000000000000000000000000000000000000000000",
    "0x000000000000000000000000e3571fe73437712481e67fed29f6a3333209bc1a",
    "0x000000000000000000000000000000000000000000000000000000000021e35"
  ],
  "data": "0x",
  "blockHash": "0xe11e84d14adda5ece90b28f4c30fd22c262ad9226e0de45d06b2edfd948a4465",
  "blockNumber": "0x18fbffb",
  "blockTimestamp": "0x6ad0cb7f",
  "transactionHash": "0xf85d4cc7fd4547836113ed0746426f7bde76dcc0351859a28b19edf30fc22ad5",
  "transactionIndex": "0x30",
  "logIndex": "0xa9",
  "removed": false
}
```

- 0x889edc2edab5f40e902b864ad4d7ade8e412f9b1
  - gets stETH
  - Mints a numbered NFT
- What else?
- **eth_getLogs** all emitted events
  - last 7 days

Transfer	Operation	Logs
`0xddf252ad` (ERC-721 `Transfer`)	mint	616
`0xddf252ad` (ERC-721 `Transfer`)	burn	665
`0xddf252ad` (ERC-721 `Transfer`)	between	2

Step 2
Watch others

topic0	Topics / data words	Logs	Txs	Senders	Main calls (tx.to / selector)
`0xddf252ad` (ERC-721 `Transfer`)	4 / 0	1,283	1,136	728	mixed
`0x6ad26c5e`	4 / 1	**665**	546	**484**	`0x889e` / `0xe3afe0a3` (456), `0x889e` / `0xf8444436` (26), others
`0xf0cb471f`	4 / 2	616	588	451	`0x889e` / `0xacf41e4d`, `0xd6681042`, `0x7951b76f`, others
`0x197874c7`	3 / 3	**8**	8	6	`0x1cff79cd` on 6 different contracts
`0x6bd5c950`	1 / 2	**8**	8	6	same 8 txs as above

- **0xe3afe0a3** the most used selector to burn the NFT
- 0x918aef2f671ff7c093f913ccc6173c56234ae040c131655e60c64dc0b143d9b9
- **debug_traceTransaction**
- Burns the NFT
- Releases ETH back to the user


```
"calls": [
  {
    "from": "0x889edc2edab5f40e902b864ad4d7ade8e412f9b1",
    "gas": "0x1381",
    "gasUsed": "0x0",
    "to": "0xba4a149c51d2dfde70c5225e085d89ab07ebc430",
    "input": "0x",
    "value": "0x2017a67f731740000",
    "type": "CALL"
  }
],
```



```
{
  "address": "0x889edc2edab5f40e902b864ad4d7ade8e412f9b1",
  "topics": [
    "0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef",
    "0x000000000000000000000000ba4a149c51d2dfde70c5225e085d89ab07ebc430",
    "0x0000000000000000000000000000000000000000000000000000000000000000",
    "0x000000000000000000000000000000000000000000000000000000000021d22"
  ],
  "data": "0x",
  "position": "0x1",
  "index": "0x1"
}
```


Step 2
Watch others

- Impersonate the user
- Approve 1 stETH
- Deposit 1 stETH
- Get the NFT
- Find the NFT ID
- Call redemption
- Get ETH

Step 3
Try it, as that user

```
❯ cast call 0x889edc2edab5f40e902b864ad4d7ade8e412f9b1 \
  --data "$(cast concat-hex 0xf8444436 "$(cast to-uint256 "139412")")" \
  --from 0xde804e46f67ebeecbe4c0a54efe33143cb7ab9e2 --rpc-url "$RPC"
```

Step 3
Try it, as that user

```
❯ cast call 0x889edc2edab5f40e902b864ad4d7ade8e412f9b1 \
  --data "$(cast concat-hex 0xf8444436 "$(cast to-uint256 "139412")")" \
  --from 0xde804e46f67ebeecbe4c0a54efe33143cb7ab9e2 --rpc-url "$RPC"
Error: server returned an error response: error code 3: execution reverted: custom error 0x4ae50228: 0000000000000000000000000000000000000000000000000000000000022094, data: "0x4ae5022800000000000000000000000000000000000000000000000000000000000022094"
```

topic0	Topics / data words	Logs	Txs	Senders	Main calls (tx.to / selector)
`0xddf252ad` (ERC-721 `Transfer`)	4 / 0	1,283	1,136	728	mixed
`0x6ad26c5e`	4 / 1	**665**	546	**484**	`0x889e` / `0xe3afe0a3` (456), `0x889e` / `0xf8444436` (26), others
`0xf0cb471f`	4 / 2	616	588	451	`0x889e` / `0xacf41e4d` , `0xd6681042` , `0x7951b76f` , others
`0x197874c7`	3 / 3	**8**	8	6	`0x1cff79cd` on 6 different contracts
`0x6bd5c950`	1 / 2	**8**	8	6	same 8 txs as above

Step 4
Prove it

- Events from both **0x197874c7** and **0x6bd5c950** are very similar
- Events all have an increasing interval
- Our NFT has ID **139412**


```
  {
    "address": "0x889edc2edab5f40e902b864ad4d7ade8e412f9b1",
    "topics": [
      "0x197874c72af6a06fb0aa4fab45fd39c7cb61ac0992159872dc3295207da7e9eb",
      "0x0000000000000000000000000000000000000000000000000000000000022002",
      "0x000000000000000000000000000000000000000000000000000000000002202e"
    ],
    "data": "0x0000000000000000000000000000000000000000000000000000000000000151bb9bc5c726dd046d0000000000000000
0010ee2c6964098ab3c6e0000000000000000000000000000000000000000000000000000000006ada003b",
    "blockHash": "0x2a457f521d4c3d15fccbf880ec9ed7f0e311c5156528c69c5a1304e79be89292",
    "blockNumber": "0x19083c1",
    "blockTimestamp": "0x6ada003b",
    "transactionHash": "0x234e3a638a79fc0945a1104667ce04420490c5e10aec1e626e482bdc73917166",
    "transactionIndex": "0x14",
    "logIndex": "0x1a",
    "removed": false
  },
```


139266

139310

- One batch a day, around 12:22 UTC
- Each batch covers a range of claim IDs
- The last batch covered up to #139310
- Ours is #139412
- Only claims inside a batch work
- Need to wait for tomorrow's batch
    - Who will send it?

Date (UTC)	Block	Sender	`0x197874c7` ids
10-15 12:21	26,197,882	`0x4d3ad7e8…`	138,692 → 138,787
10-16 12:23	26,205,073	`0x2ae828f1…`	138,788 → 138,823
10-17 12:22	26,212,244	`0x2ae828f1…`	138,824 → 138,919
10-18 12:21	26,219,417	`0xc1025896…`	138,920 → 138,991
10-19 12:23	26,226,596	`0x82a821e8…`	138,992 → 139,094
10-20 12:22	26,233,767	`0x82a821e8…`	139,095 → 139,196
10-21 12:21	26,240,942	`0x7abc999c…`	139,197 → 139,265
10-22 12:23	26,248,129	`0x565f04cb…`	139,266 → 139,310

Step 4
Prove it

Step 4
Prove it

- Let's test it
- Fork on **26248128**
  - One block before the last known batch

- Try to cash out **#139267** -> **Fail**
- A random address sends the batch -> **Fail**
- Usual sender sends the same batch -> **accepted**
- Cash the claim, after the real batch -> **accepted**

```
❯ cast call 0x4e3f2deeb59eb9a205d82d17647b3e56422e0fee \
        "$(cast tx 0x234e3a638a79fc0945a1104667ce04420490c5e10aec1e626e482bdc73917166 input)" \
    --from 0x1234567890123456789012345678901234567890 \
        --block 26248128
Error: server returned an error response: error code 3: execution reverted, data: "0x1db3b859"
~ ❯ _
```



Blocked

So close, yet so far

Privileged batch senders

~1 hour of expert work to get here

Full results

Blocked

- Lido
- ether.fi
- Ethena
- Maple
- Falcon Finance
- Kelp

Partial Exit

- Aave V3*
- Spark*

Hard, but Possible

- Aave V3
- Spark
- Uniswap V3
- Curve
- Pendle

Full Exit

- Morpho
- Sky
- Rocket Pool
- Compound
- Fluid

https://github.com/d6o/DefiWalkawayTest

Recommendations

- Add a permissionless fallback to every privileged step
  - If no batch, unlock or report arrives for N days, let anyone finish it
- Use standard exit functions, and make them work
  - withdraw / redeem on the position token
- One call, no traps
  - No "send, then call"; no calls that succeed and pay 0
- Run the walkaway test before launch

For users

- Ask: who has to act before I can leave?
- Check liquidity on hand, not just TVL
- Keep your own deposit transaction
- Be careful with scam tokens
- To leave alone, you need an archive node with tracing



THANK YOU!

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